

Root Finding Comparison Report

Numerical Methods Portfolio Project

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Developed Using

MATLAB Online

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Project Goal

Compare all five methods for solving:

$$f(x) = x^3 - x - 2$$

and answer:

- Which is fastest?
- Which is most reliable?
- Which is easiest to implement?
- Which should an engineer choose?

1 PROJECT OVERVIEW

This project compares five numerical root-finding methods implemented in MATLAB. The methods were applied to the nonlinear equation $x^3 - x - 2 = 0$ and evaluated based on convergence speed, computational efficiency, robustness, and implementation requirements.

2 METHODS COMPARED

1. Bisection
2. Newton-Raphson
3. Secant
4. False Position
5. Fixed Point Iteration

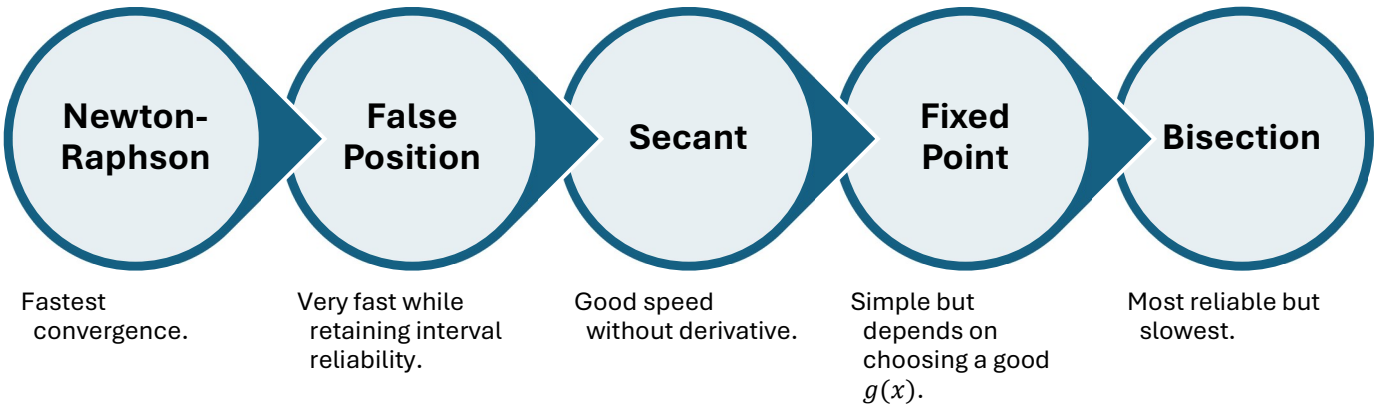
3 RESULTS TABLE

Method	Root	Iterations	Derivative Required
Bisection	1.521380	20	No
Newton-Raphson	1.521380	3	Yes
Secant	1.521380	7	No
False Position	1.521380	5	No
Fixed Point	1.521380	7	No

4 PERFORMANCE RANKING

Rank	Method
 1	Newton-Raphson
 2	False Position
 3	Secant
4	Fixed Point
5	Bisection

4.1 REASONING:



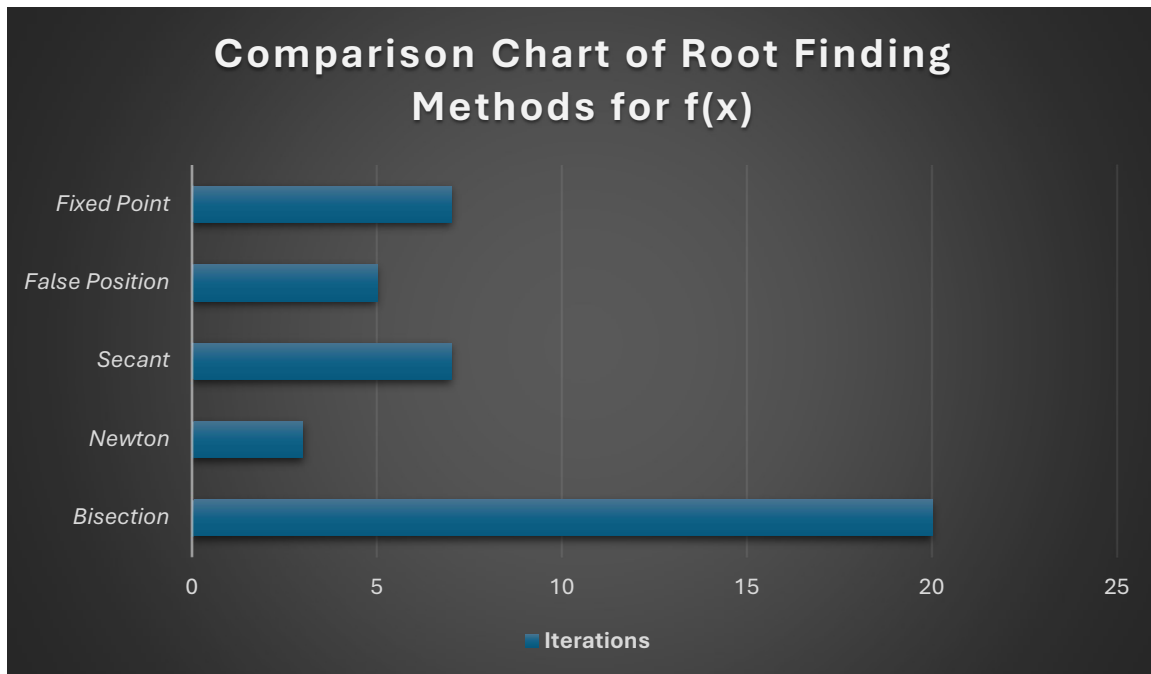
5 ENGINEERING DISCUSSION

Criterion	Best Method
Fastest Convergence	Newton-Raphson
Most Reliable	Bisection
Best Balance	False Position
No Derivative Needed	Secant
Educational Value	Fixed Point

6 ENGINEERING RECOMMENDATIONS

If derivative available	If derivative unavailable	If guaranteed convergence needed	If studying convergence behavior
Newton-Raphson	Secant	Bisection False Position	Fixed Point Iteration

7 CONVERGENCE CHART



7.1 KEY LESSONS

- Faster methods are not always the most reliable.
- Derivative information can significantly improve convergence speed.
- Interval-based methods provide guaranteed convergence.
- The choice of iteration function determines the success of Fixed Point Iteration.
- Engineering method selection depends on problem requirements.

This is exactly the kind of reflection that looks good in a portfolio.

8 CONCLUSION

All five numerical methods successfully converged to the root $x \approx 1.521380$. Newton-Raphson achieved the fastest convergence, requiring only 3 iterations, while Bisection provided the highest reliability at the expense of speed. False Position offered the best balance between convergence rate and robustness, whereas Secant eliminated the need for derivative evaluation. Fixed Point Iteration provided valuable insight into convergence behavior and the importance of selecting an appropriate iteration function.